MULTILINGUAL MULTIMODAL TOURISM AGENT FOR SRI LANKA USING KNOWLEDGE GRAPH REASONING

Project Id: 25-26J-119

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor: Date: 29/08/23

ABSTRACT

Tourism in Sri Lanka faces significant communication barriers due to the multilingual nature of the population and the prevalence of Sinhala-English code-mixing in daily conversations. This research proposes the development of a Multimodal Sinhala–English Code-Mixed Speech and Text Processing Module as part of a comprehensive MCP-enabled tourism assistant system.
The module integrates three core technologies to address unique linguistic challenges in Sri Lankan tourism. The Speech-to-Text (STT) component employs fine-tuned multilingual models trained on custom Sinhala-English code-mixed datasets, processing natural code-mixed expressions commonly used in tourist interactions.

The Text-to-Speech (TTS) system incorporates advanced emotion detection and regional accent adaptation capabilities to provide contextually appropriate responses that reflect the natural speech patterns of local communities. This ensures tourists receive authentic, culturally sensitive audio feedback that enhances their overall experience.

The Vision-based Optical Character Recognition (OCR) module leverages existing APIs enhanced with custom Sinhala script processing and transliteration algorithms. This component converts signboard text into accessible Singlish format with optional audio output, enabling tourists to understand local signage, menus, and informational displays effortlessly.

The comprehensive system processes natural code-mixed expressions that occur frequently in real-world tourist scenarios, bridging the communication gap between international visitors and local service providers. By combining these multimodal technologies, the platform creates a seamless interface that supports both spoken and written communication needs.

This innovative approach addresses the practical challenges of multilingual tourism environments while preserving the cultural authenticity of local language use patterns, ultimately enhancing visitor satisfaction and supporting the growth of Sri Lanka's tourism industry.

Keywords: Code-mixing, Speech Processing, Multimodal AI, Tourism Technology, Sinhala Language Processing

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ACM	Association for Computing Machinery
AI	Artificial Intelligence
API	Application Programming Interface
ASR	Automatic Speech Recognition
CER	Character Error Rate
CNN	Convolutional Neural Network
DSR	Design Science Research
FR	Functional Requirement
GPU	Graphics Processing Unit
LLM	Large Language Model
ML	Machine Learning
MOS	Mean Opinion Score
NFR	Non-Functional Requirement
NLP	Natural Language Processing
OCR	Optical Character Recognition
OpenAPI	Open Application Programming Interface
REST	Representational State Transfer
RNN	Recurrent Neural Network
RTF	Real-time Factor
SDK	Software Development Kit
SQL	Structured Query Language
STT	Speech-To-Text
TTS	Text-to-Speech
WER	Word Error Rate

Table 1 - List of Abbreviation

1. INTRODUCTION

The tourism industry in Sri Lanka operates within a uniquely complex multilingual environment where Sinhala, Tamil, and English languages intersect in intricate patterns of daily communication. This linguistic diversity, while representing the rich cultural heritage of the island nation, presents substantial challenges for effective communication between tourists and local service providers. The phenomenon of code-mixing and code-switching, where speakers seamlessly alternate between multiple languages within single conversations or even individual sentences, has become increasingly prevalent in tourist-local interactions across hospitality, transportation, retail, and recreational sectors.

Sri Lanka's position as a prominent South Asian tourist destination has experienced remarkable growth over the past decade, attracting visitors from diverse linguistic backgrounds including English, German, French, Chinese, Japanese, and various other languages. However, the communication barriers arising from the country's multilingual landscape significantly impact service quality delivery and overall visitor satisfaction. Local service providers, particularly in rural and semi-urban tourist destinations, often struggle to communicate effectively with international visitors while simultaneously managing their natural tendency to code-mix between Sinhala and English during service interactions.

Traditional language processing technologies have predominantly focused on monolingual or simple bilingual communication scenarios, with limited attention to the complex code-mixing patterns characteristic of South Asian linguistic behaviors. Existing speech recognition systems, text-to-speech applications, and optical character recognition tools demonstrate poor performance when processing code-mixed content, particularly in the Sinhala-English language pair. This technological gap creates a critical need for specialized processing systems capable of understanding, interpreting, and generating

code-mixed speech and text content that reflects authentic Sri Lankan communication patterns.

The integration of modern Natural Language Processing techniques, Automatic Speech Recognition systems, Text-to-Speech synthesis, and computer vision technologies presents unprecedented opportunities to address these multilingual communication challenges. Recent developments in Large Language Models, Convolutional Neural Networks, and Recurrent Neural Networks have demonstrated promising capabilities in handling complex linguistic patterns, though their application to South Asian code-mixing scenarios remains largely unexplored.

This research addresses the critical gap in multilingual speech and text processing technologies by developing a comprehensive Sinhala-English code-mixed processing system integrated with Speech-to-Text conversion, Text-to-Speech synthesis, and vision-based Sinhala Optical Character Recognition capabilities with transliteration functionality. The proposed system aims to enhance tourism communication experiences by providing real-time processing of naturally occurring code-mixed interactions, thereby improving service delivery quality and visitor satisfaction across Sri Lanka's diverse tourism landscape.

The significance of this research extends beyond tourism applications, offering potential benefits to education, healthcare, business communications, and digital content creation sectors where Sinhala-English code-mixing patterns are increasingly prevalent. By addressing the technological challenges associated with multilingual speech and text processing in the Sri Lankan context, this study contributes to the broader understanding of South Asian computational linguistics while providing practical solutions for immediate implementation in tourism and related industries.

1.1. Background and Literature Survey

The tourism industry in Sri Lanka encounters significant communication challenges due to the country's multilingual landscape where Sinhala, Tamil, and English coexist in complex patterns of usage. Code-mixing and code-switching phenomena are particularly prevalent in tourist-local interactions, where speakers naturally alternate between languages within single conversations or even individual sentences. This linguistic complexity presents substantial barriers to effective tourism service delivery and visitor experience quality.

Recent advances in Natural Language Processing (NLP) and speech technologies have opened new possibilities for addressing multilingual communication challenges. However, existing solutions primarily focus on monolingual or simple bilingual scenarios, with limited research addressing the specific patterns of South Asian code-mixing behaviors. The emergence of transformer-based architectures and large language models has demonstrated promising capabilities in multilingual understanding, yet their application to low-resource language pairs like Sinhala-English remains underexplored.

1.1.1. Code-Mixing and Code-Switching Research

Code-mixing research has evolved significantly from early sociolinguistic studies to computational approaches. As defined in [1], code-mixing is “the mixing of various linguistic units primarily from two participating grammatical systems within a sentence”. This differs from code-switching, which involves alternation between languages at sentence or discourse boundaries. In the Sri Lankan context, code-mixing patterns follow unique grammatical and social conventions that differ from other documented language pairs.

Recent computational studies have identified three primary code-mixing processes: insertion (embedding lexical items from one language into another), alternation (switching

between languages at clause boundaries), and congruent lexicalization (using shared grammatical structures).

1.1.2. Speech Processing for Low-Resource Languages

Speech processing research for Sinhala has made substantial progress with the availability of Google's multi-speaker Sinhala speech corpus and the 185k-utterance Sinhala ASR dataset through OpenSLR. However, code-mixed speech resources remain severely limited. Traditional approaches have relied on separate language identification followed by language-specific processing, but this pipeline approach often fails with rapid code-switching patterns common in natural Sri Lankan speech.

Modern end-to-end approaches using architectures like Whisper (OpenAI), Wav2Vec 2.0 (Facebook), and ESPnet have demonstrated superior performance on multilingual tasks.

1.1.3. Text-to-Speech Synthesis Advances

Contemporary TTS research has transitioned from concatenative and parametric approaches to neural synthesis methods. Autoregressive models like Tacotron 2 provide high-quality synthesis but suffer from slow inference speeds. Non-autoregressive alternatives such as FastSpeech 2 and VITS offer real-time generation capabilities essential for interactive applications.

Emotion-aware TTS synthesis has emerged as a critical requirement for natural human-computer interaction. Recent work demonstrates controllable emotional synthesis through explicit emotion conditioning and speaker adaptation techniques. However, research specifically addressing accent adaptation for Sri Lankan English varieties and emotional expression patterns in Sinhala speech remains limited.

1.1.4. Multimodal Integration and OCR Technology

Vision-based text recognition has achieved remarkable accuracy improvements through deep learning approaches. Transformer-based architectures and CNN-RNN hybrid models

have largely superseded traditional feature-based methods. For Sinhala script recognition, Tesseract OCR provides basic capabilities, but challenges remain with complex conjunct characters, contextual variations, and real-world image conditions commonly encountered in tourism scenarios.

Transliteration research for Sinhala-Singlish conversion has primarily focused on rule-based approaches, though recent neural methods show promise for handling phonetic variations and contextual adaptations. The integration of OCR output with speech synthesis for accessibility purposes represents an underexplored area with significant potential impact.

1.2.Research Gap

Despite significant advances in multilingual NLP and speech processing, several critical gaps exist in the current research landscape for code-mixed speech and text processing in tourism applications. These gaps are discussed below with reference to existing literature.

Gap 1: Code-Mixed Speech Processing for Tourism Context

Most existing research predominantly focuses on high-resource language pairs such as English–Hindi and English–Spanish, which are the most represented in code-mixed datasets and benchmarks. In contrast, Sinhala–English code-mixing appears in only a handful of studies, highlighting its status as an underexplored low-resource language pair [3]. While Shashikala (2021) explored sociolinguistic aspects of Sinhala-English code-mixing among urban bilingual speakers, the focus remained on structural properties rather than speech processing applications [4]. Silva et al. (2019) noted challenges in processing Sinhala-English code-switching and code-mixing in chatbot development platforms, highlighting the gap in practical applications [5]. Recent computational work has primarily addressed text-based code-mixed data for hate speech detection rather than speech processing for tourism contexts [6], [7]. Tourism-specific vocabulary, cultural expressions, and contextual usage patterns remain unaddressed in current speech

processing systems, particularly for the unique grammatical and social conventions that characterize Sri Lankan code-mixing patterns.

Gap 2: Integrated Multimodal Processing

While individual components (STT, TTS, OCR) have received substantial research attention, integrated systems that seamlessly combine these modalities for specific domain applications are rare. Recent research by Prabhu et al. (2025) identified key challenges in multimodal deployment including "increased sensory requirements, computational cost, and modality synchronization" that constrain real-world applications [8]. While TourMLLM (2025) addressed tourism domain applications, it focused on textual retrieval-augmented generation rather than speech and vision integration [9]. The tourism domain presents unique requirements for real-time processing, emotional appropriateness, and cultural sensitivity that existing multimodal systems do not adequately address. Current approaches typically process modalities in isolation rather than leveraging their complementary nature for enhanced understanding of tourist-local interactions.

Gap 3: Accent and Emotion Adaptation

Current TTS systems lack adequate support for Sri Lankan accent variations and culturally appropriate emotional expression patterns. Existing Sinhala speech recognition research has focused on basic corpus development without addressing accent variations or emotional expression patterns [10]. While recent work by Zhang et al. (2023) and Rathnayake et al. (2023) has shown promising results with adapter-based approaches for multilingual tasks, research specifically addressing accent adaptation for Sri Lankan English varieties remains limited [11], [12]. The relationship between linguistic choice (Sinhala vs. English vs. code-mixed) and appropriate prosodic patterns in tourism contexts remains unexplored. This gap is particularly significant given that emotional appropriateness and cultural sensitivity are crucial factors in tourism service delivery.

1.3. Research Problem

The primary research problem addressed by this study is:

How can a multimodal speech and text processing system effectively handle Sinhala-English code-mixing patterns in tourism contexts while providing real-time, emotionally appropriate, and culturally sensitive responses through integrated STT, TTS, and OCR capabilities?

This overarching problem encompasses several specific technical challenges:

1. **Code-Mixed Speech Recognition Challenge**: Achieving acceptable Word Error Rates for spontaneous Sinhala-English code-mixed speech in noisy tourism environments.
2. **Contextual TTS Synthesis Challenge**: Generating natural, accent-appropriate, and emotionally suitable speech output that maintains linguistic authenticity across code-mixed content.
3. **Real-Time Multimodal Integration Challenge**: Seamlessly combining speech, text, and visual input modalities while maintaining response latencies suitable for interactive tourism applications.
4. **Cultural and Linguistic Appropriateness Challenge**: Ensuring system responses align with Sri Lankan cultural norms and linguistic expectations across diverse user demographics.

2. OBJECTIVES

2.1. Main Objective

To develop and evaluate a comprehensive Multimodal Sinhala–English Code-Mixed Speech and Text Processing Module that enables natural, culturally appropriate, and real-time multilingual communication for Sri Lankan tourism applications through integrated Speech-to-Text, Text-to-Speech, and Optical Character Recognition capabilities.

2.2. Specific Objectives

Objective 1: Code-Mixed Speech Recognition System Development

This objective focuses on creating a specialized Speech-to-Text system engineered to handle the unique challenges of Sinhala-English code-mixed utterances prevalent in Sri Lankan tourism contexts. The system will employ state-of-the-art transformer-based architectures fine-tuned specifically for code-mixed speech patterns, addressing the limitations of existing monolingual approaches that fail to accurately process intra-sentential language switching. The development process includes creating and annotating a comprehensive dataset of over 10 hours of code-mixed speech data, ensuring representation of diverse Sri Lankan accents, regional variations, and tourism-specific vocabulary. This dataset will encompass various tourist interaction scenarios including hotel bookings, restaurant orders, transportation inquiries, and cultural site visits, providing the foundation for training robust models that can handle real-world communication patterns between tourists and local service providers.

- Develop a robust Speech-to-Text system capable of accurately transcribing Sinhala-English code-mixed utterances
- Create and annotate a comprehensive dataset of 10 hours+ of code-mixed speech data representing diverse Sri Lankan accents and tourism contexts

Objective 2: Emotion-Aware TTS System Implementation

This objective aims to design and implement an advanced Text-to-Speech system that supports seamless synthesis across Sinhala, English, and code-mixed input while maintaining natural prosody and emotional appropriateness. The system will incorporate emotion detection capabilities to analyze input text sentiment and context, enabling the generation of speech output with appropriate emotional coloring that enhances user experience and cultural sensitivity. Regional accent adaptation features will ensure that the synthesized speech reflects authentic Sri Lankan pronunciation patterns and intonation characteristics. The implementation will address the challenge of maintaining linguistic coherence across language boundaries within code-mixed utterances while preserving the natural flow and rhythm that characterizes authentic Sri Lankan multilingual speech patterns in tourism interactions.

- Design and implement a Text-to-Speech system supporting Sinhala, English, and code-mixed input with natural prosody generation

Objective 3: Vision-Based OCR and Transliteration Module

This objective encompasses the development of a comprehensive visual text processing system that leverages existing OCR APIs enhanced with custom Sinhala script recognition capabilities. The module will process real-world visual text from signboards, menus, brochures, and other tourism-related materials, converting Sinhala text into accessible digital format. A critical component involves implementing intelligent Sinhala-to-Singlish transliteration algorithms that can handle common place names, tourism terminology, and cultural references, making them accessible to international visitors. The system will incorporate contextual understanding to provide appropriate transliterations based on tourism contexts, ensuring that cultural nuances and local naming conventions are preserved while maintaining readability for non-Sinhala speakers. Integration with the TTS system will enable audio output of transliterated content, support accessibility and enhancing user experience.

- Develop a Sinhala OCR system leveraging existing API

- Implement Sinhala-to-Singlish transliteration for common place names and tourism-related terms

Objective 4: Evaluation and Performance Optimization

This objective establishes a comprehensive evaluation framework to assess system performance across multiple dimensions and optimize functionality for real-world deployment. Standard speech processing metrics including Word Error Rate (WER) and Character Error Rate (CER) will be employed alongside Mean Opinion Score (MOS) evaluations to measure user perception of speech quality and naturalness. Domain-specific evaluation metrics will focus on tourism task completion rates, measuring the system's effectiveness in facilitating actual tourist-local interactions such as booking confirmations, navigation assistance, and cultural information exchange. The evaluation process will include user studies with both tourists and local service providers to assess practical utility, cultural appropriateness, and overall satisfaction. Performance optimization will address latency requirements for mobile applications, accuracy improvements through iterative model refinement, and resource efficiency considerations essential for deployment in diverse technological environments across Sri Lanka's tourism infrastructure.

- Conduct comprehensive evaluation using standard metrics (WER, CER, MOS) and domain-specific tourism task completion rates

3. METHODOLOGY

This chapter presents the methodology for developing the Multimodal Sinhala-English Code-Mixed Speech and Text Processing Module for tourism applications. The approach follows a Design Science Research framework with experimental evaluation, incorporating iterative cycles of design, implementation, testing, and refinement through both quantitative performance metrics and qualitative user experience assessments. The methodology integrates theoretical foundations from computational linguistics, speech processing, and human-computer interaction to address code-mixed speech processing complexities while meeting practical deployment requirements for real-world tourism applications in Sri Lanka.

3.1.System Architecture and Design

3.1.1. Overall System Architecture

The proposed system follows a microservices architecture pattern with clear separation of concerns:

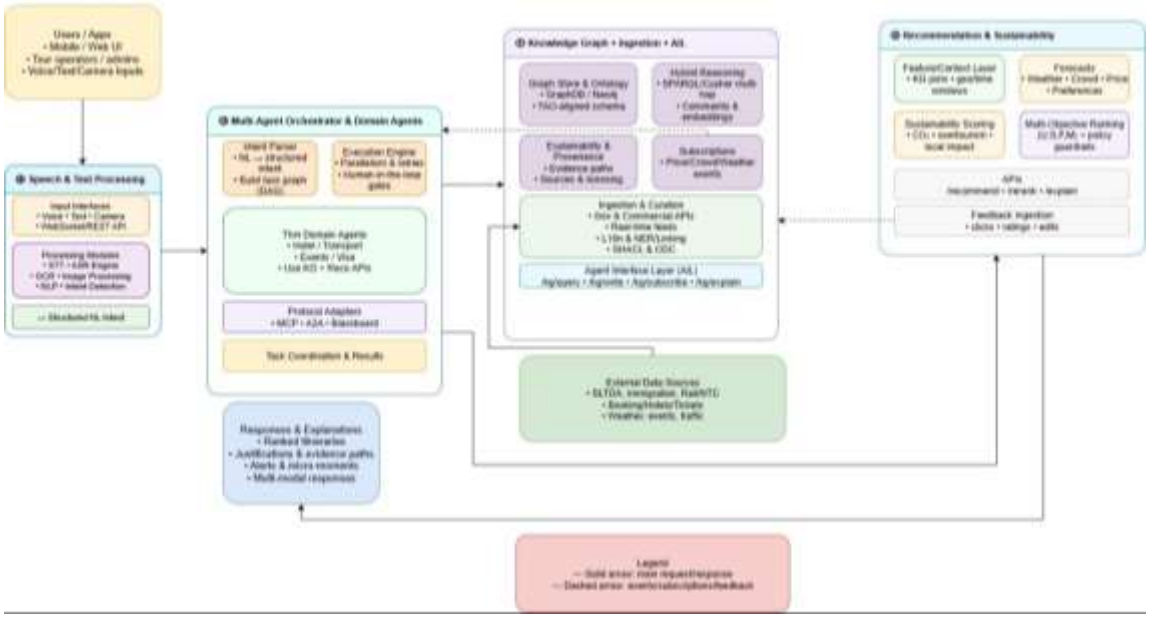

Figure 1 - Overall System Diagram

3.1.2. Component Overview

Mobile Application Layer – The user-facing app (voice, text, camera). It captures speech or images and displays responses (audio/text).

API Gateway & Load Balancer – The single entry point. It authenticates requests, rate-limits, logs, and routes traffic to healthy services so the system scales reliably.

Speech & Text Processing Service – The core runtime that orchestrates three specialized modules:

- **STT Module (Speech-to-Text):** Real-time recognition for Sinhala–English code-mixed speech, with multilingual handling and accent adaptation to improve accuracy in local contexts.
- **TTS Module (Text-to-Speech):** Synthesizes natural audio with optional **emotional tone**, voice presets, accent adaptation, and **prosody control** for clear, human-like delivery.
- **OCR Module:** Detects Sinhala text in images, performs **Singlish transliteration**, applies image enhancement for better readability, and can return audio output for read-aloud.

Data & Model Layer – Central store for training data, acoustic/linguistic models, OCR/TTS assets, configs, and logs. It supports continuous improvement (fine-tuning, feedback, monitoring) and feeds updated models back to the service.

Flow: App → Gateway → Processing Service → (STT/TTS/OCR as needed) → results returned to the app, with artifacts and telemetry persisted in the Data & Model Layer. This setup cleanly separates concerns, scales horizontally, and supports real-time, multilingual, code-mixed interactions.

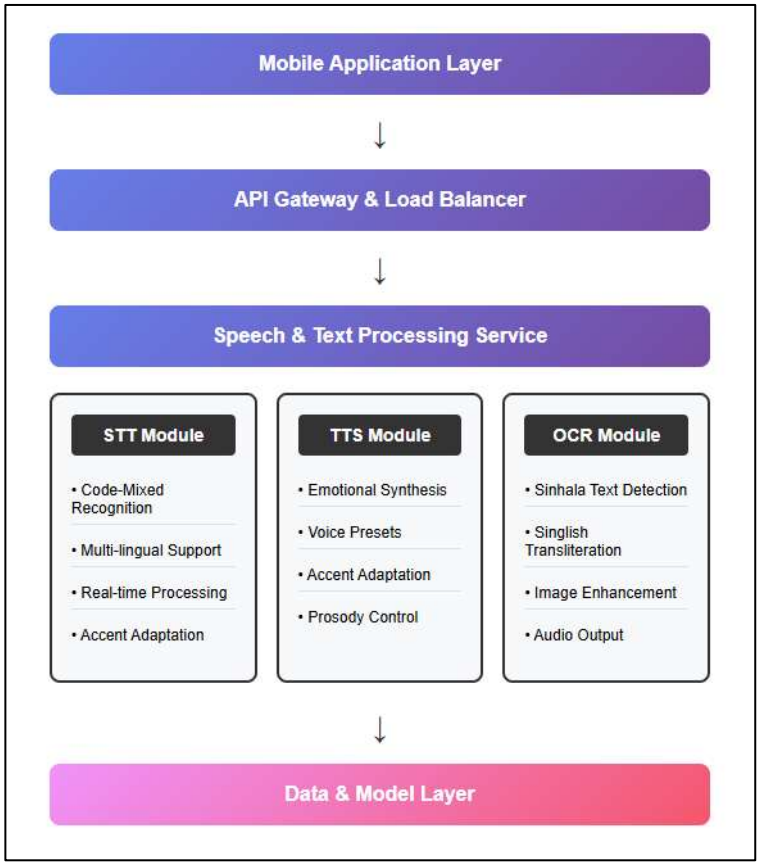

Figure 2 - Component Diagram

3.1.3. Technology Stack Selection

Backend Services:

- **Framework**: FastAPI (Python) for REST APIs with automatic OpenAPI documentation
- **ML Models**: PyTorch for custom model development and fine-tuning
- **Audio Processing**: librosa, soundfile, webrtcvad for audio preprocessing and feature extraction
- **Image Processing**: OpenCV, PIL for image preprocessing and enhancement
- **Database**: PostgreSQL for metadata storage, Redis for caching and session management

- **Message Queue**: Kafka for asynchronous task processing

Infrastructure:

- **Containerization**: Docker containers with Kubernetes orchestration
- **Cloud Platform**: Azure
- **API Management**: Kong API Gateway with rate limiting and authentication
- **Monitoring**: OpenTelemetry + Prometheus + Grafana for observability

3.2. Data Collection and Preparation Strategy

3.2.1. Speech Data Collection

Corpus Design

- Systematic collection of code-mixed speech data from tourism contexts
- Recording sessions at major tourist locations (airports, hotels, cultural sites)
- Collaboration with tourism industry partners for authentic conversation samples
- Target: 8 hours of diverse code-mixed speech across different scenarios

Data Sources:

- **Tourist-Guide Interactions**: Recorded conversations at heritage sites and tourist attractions
- **Hotel and Restaurant Interactions**: Service conversations with code-mixing patterns
- **Transportation Context**: Taxi, tuk-tuk, and public transport conversations
- **Shopping and Market Interactions**: Negotiation and inquiry conversations
- **Emergency and Information Seeking**: Help-seeking conversations with authorities

3.3. Evaluation Methodology

3.3.1. Quantitative Evaluation Metrics

Speech-to-Text Evaluation:

- **Word Error Rate (WER)**: Standard metric for transcription accuracy
- **Character Error Rate (CER)**: Fine-grained accuracy measurement
- **Real-time Factor (RTF)**: Processing speed relative to audio duration
- **Language Identification Accuracy**: Correct identification of code-mixing patterns

Text-to-Speech Evaluation:

- **Mean Opinion Score (MOS)**: Subjective quality assessment (1-5 scale)
- **Naturalness Rating**: Perceived human-likeness of synthesized speech
- **Emotion Accuracy**: Correctness of emotional expression
- **Accent Appropriateness**: Cultural and regional authenticity

OCR and Transliteration Evaluation:

- **Character Recognition Accuracy**: Correct Sinhala character identification
- **Word-level Accuracy**: Complete word recognition rate
- **Transliteration Accuracy**: Correct Sinhala-to-Singlish conversion
- **End-to-end Task Completion**: Successful signboard reading and audio output

3.3.2. Qualitative Evaluation Approach

Evaluation Scenarios:

- **Information Seeking**: Asking for directions, recommendations, information
- **Service Interactions**: Hotel booking, restaurant orders, transportation
- **Emergency Situations**: Help-seeking, medical assistance, reporting problems
- **Cultural Exploration**: Learning about traditions, history, customs

3.4. Component Integration and Data Flow

The system achieves the main objective through the following integrated workflow:

1. Multimodal Input Processing:

- Voice input captured via mobile microphone → STT Module
- Text input from keyboard/chat → Direct text processing
- Camera input from signboard scanning → OCR Module

2. Language Processing Pipeline:

- Language detection identifies Sinhala, English, or code-mixed content
- Appropriate processing pathway selected based on input modality and language
- Context preservation across modalities for coherent responses

3. Intelligent Response Generation:

- Processed input integrated with tourism knowledge graph (Component 2)
- Multi-agent orchestrator (Component 3) coordinates response planning
- Personalization engine (Component 4) tailors response to user preferences

4. Multimodal Output Generation:

- Text responses generated with cultural and contextual appropriateness
- TTS converts text to speech with appropriate accent and emotion
- Visual elements (transliterated text, images) provided when relevant

3.5. Materials and Equipment Requirements

Hardware Requirements:

- High-performance development workstation with GPU support
- Professional audio recording equipment (microphones, interfaces)
- Mobile testing devices (Android/iOS across different specifications)
- Storage infrastructure for large datasets

Software Requirements:

- Machine learning frameworks (PyTorch, TensorFlow, HuggingFace)
- Cloud computing resources (AWS GPU instances)
- Development tools (IDEs, version control, collaboration platforms)
- Specialized software for audio processing and computer vision

Cloud Infrastructure:

- GPU compute instances for model training
- Object storage for datasets and model artifacts
- API services for OCR and comparative testing
- Monitoring and logging infrastructure

3.6.System Requirements Specification

3.6.1. Functional Requirements

FR1: Speech-to-Text Processing

- FR1.1: The system SHALL accurately transcribe Sinhala speech
- FR1.2: The system SHALL accurately transcribe English speech
- FR1.3: The system SHALL process code-mixed Sinhala-English speech
- FR1.4: The system SHALL detect and tag language switches within utterances
- FR1.5: The system SHALL support real-time streaming speech recognition

FR2: Text-to-Speech Synthesis

- FR2.1: The system SHALL synthesize natural Sinhala speech from text input
- FR2.2: The system SHALL synthesize natural English speech with Sri Lankan accent
- FR2.3: The system SHALL handle code-mixed text input for speech generation
- FR2.4: The system SHALL support emotion control in synthesized speech
- FR2.5: The system SHALL provide multiple voice presets for different contexts

FR3: Optical Character Recognition

- FR3.1: The system SHALL detect Sinhala text in real-world images
- FR3.2: The system SHALL handle various image conditions (lighting, perspective, occlusion)
- FR3.3: The system SHALL transliterate Sinhala text to Singlish
- FR3.4: The system SHALL provide audio output of detected and transliterated text

FR4: API and Integration

- FR4.1: The system SHALL provide RESTful APIs for all major functions
- FR4.2: The system SHALL support real-time data streaming
- FR4.3: The system SHALL integrate with mobile applications
- FR4.4: The system SHALL provide error handling and graceful degradation

3.6.2. Non-Functional Requirements

NFR1: Performance

- NFR1.1: The system SHALL process speech-to-text requests in near real-time
- NFR1.2: The system SHALL provide responsive text-to-speech synthesis
- NFR1.3: The system SHALL handle concurrent user requests efficiently

NFR2: Reliability

- NFR2.1: The system SHALL maintain high availability during operational hours
- NFR2.2: The system SHALL provide consistent results for identical inputs
- NFR2.3: The system SHALL recover gracefully from processing failures

NFR3: Security

- NFR3.1: All communications SHALL use secure encryption protocols
- NFR3.2: User audio and image data SHALL be processed securely
- NFR3.3: The system SHALL implement appropriate authentication mechanisms

NFR4: Usability

- NFR4.1: The system SHALL provide intuitive interfaces for mobile integration
- NFR4.2: The system SHALL deliver clear and actionable error messages
- NFR4.3: The system SHALL support basic offline functionality where applicable

4. DESCRIPTION OF PERSONNEL AND FACILITIES

Task Name	Progress	Mar	Apr	May	Jun	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
Project Initiation																	
Study about research gaps	100%																
Deciding appropriate research area	100%																
Going through research papers	100%																
Finalizing a topic	100%																
Requirement Gathering & Analysis																	
Identifying functional requirements	100%																
Identifying non-functional requirements	100%																
Identifying data requirements	100%																
Identifying knowledge requirements	100%																
Design																	
Creating the Figma prototype	0%																
Development																	
Knowledge Graph Creation	0%																
Backend Development	0%																
Frontend Development	0%																
Testing																	
Unit and API testing	0%																
Integrating Testing	0%																
Load testing	0%																
User Acceptance Testing	0%																
Deployment	0%																

Table 2 - Gantt Chart

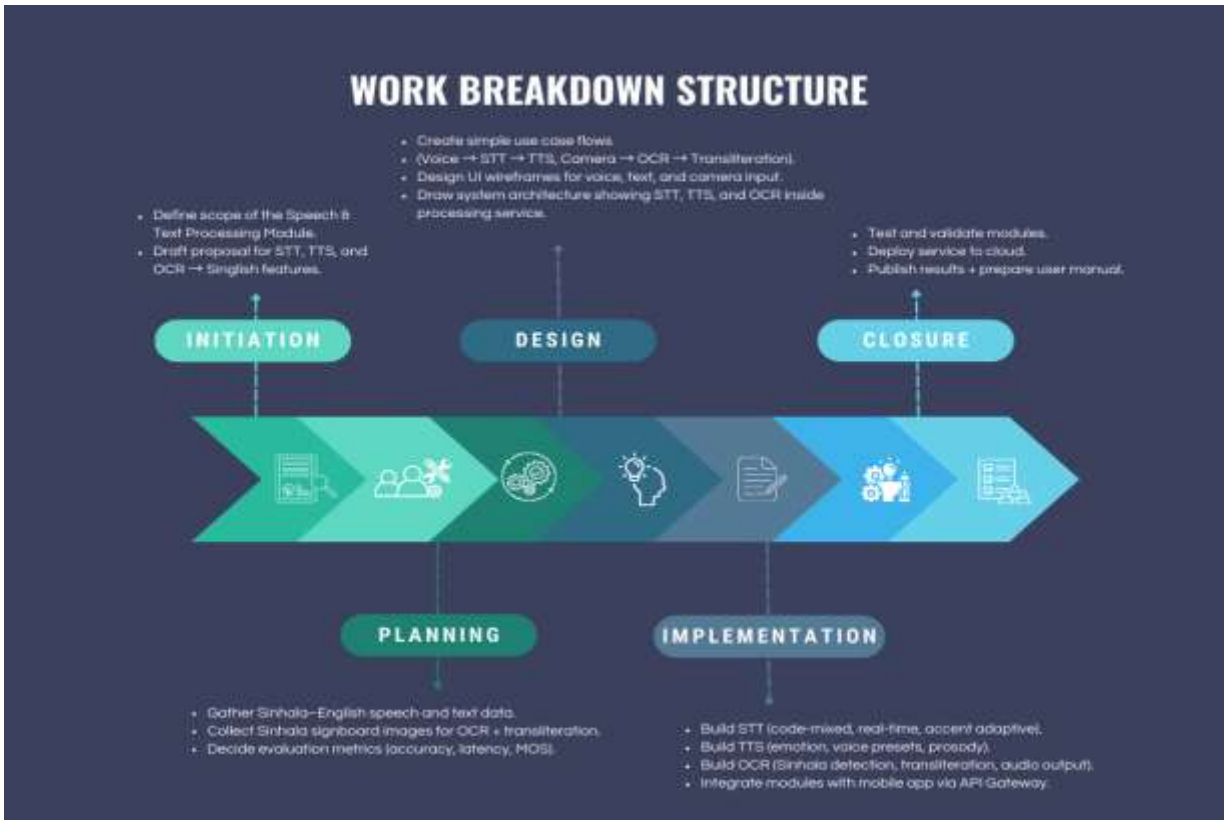

Figure 3 - Work Breakdown Structure

5. BUDGET AND BUDGET JUSTIFICATION

The proposed system has been planned with a lean but realistic budget, making use of free tiers where possible, while allocating essential costs for cloud services, APIs, and mobile deployment. The emphasis is on using **Azure cloud services** for backend hosting, storage, and speech capabilities, combined with **OpenAI API keys** for multi-agent reasoning and natural language tasks. In addition, a **Flutter-based mobile application** will be developed as the final deliverable, requiring costs for developer accounts on app stores. Hardware requirements such as microphones are also included to support data collection and testing. A buffer allocation has been added to account for unforeseen expenses during integration and field validation.

Description	Estimated Cost	Notes
Azure Cloud Services (VM hosting, Blob Storage, App Services, Speech-to-Text)	LKR 3000 / monthly (with free tier credits initially)	Backend hosting, data storage, STT/OCR services
OpenAI API Keys	LKR 3000	For embeddings, orchestration, reasoning tasks
Microphones for Voice Data Collection	LKR 10,000	For Sinhala–English code-mixed dataset collection
Mobile Testing Device	LKR 20,000	For app testing and evaluation
Miscellaneous / Buffer		Documentation, SD cards, adapters, printing

Table 3 - Budget Table

5.1. Budget Justification

Azure Cloud Services: Azure has been selected as the primary cloud provider due to its generous free tiers and reliable integration of storage, compute, and AI services. Virtual Machines and App Services will host the backend and APIs, while Blob Storage will hold datasets and graph exports. Azure Speech-to-Text will support Sinhala–English transcription at competitive rates, making it essential for the multilingual component.

OpenAI API Keys: A developer-tier plan will be used for embedding generation, agent orchestration, and conversational reasoning tasks. This ensures that the system can perform advanced semantic queries and multi-agent coordination effectively.

Flutter Mobile Application: The final deliverable will be a cross-platform mobile application built using Flutter, allowing deployment on both Android and iOS. Publishing costs include the Apple Developer Program ($99/year) and Google Play Developer registration ($25 one-time). These are unavoidable costs to ensure the app can be distributed officially and maintained with regular updates.

Microphones & Mobile Device: A high-quality microphone (~LKR 10,000) is required for collecting Sinhala–English speech data to fine-tune STT components. Additionally, a mid-range Android device (~LKR 20,000) is budgeted for testing the mobile application in real-world conditions.

Miscellaneous Costs: A buffer of $100 is included for documentation, printing, SD cards, adapters, and other unforeseen requirements during development and evaluation.

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